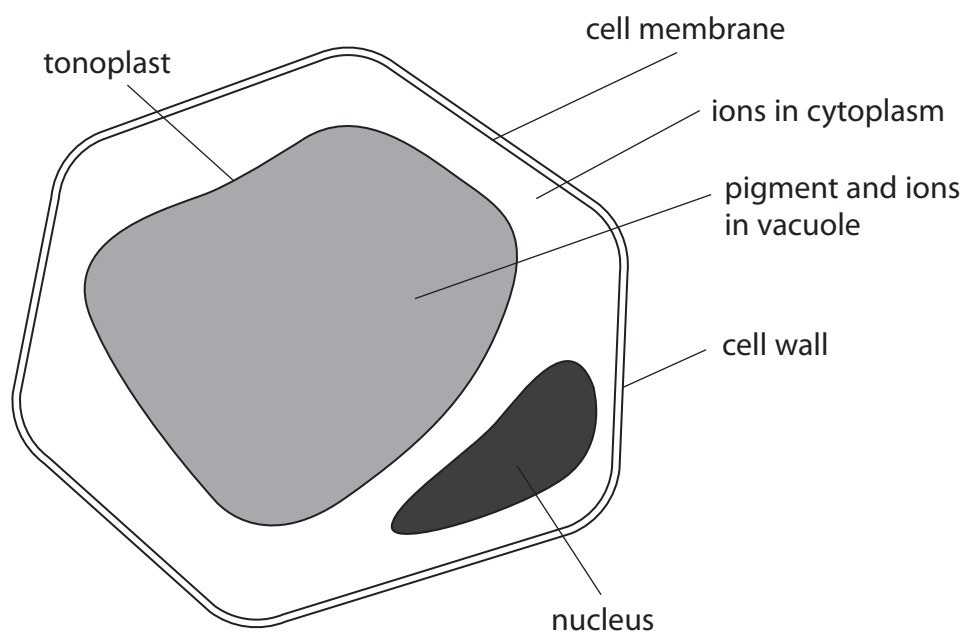


2 Plant cells have a cell membrane and a tonoplast.

The diagram shows these structures and the location of pigment and ions.



- (a) Name **three** membrane-bound structures found in some plant cells that are not shown in the diagram.

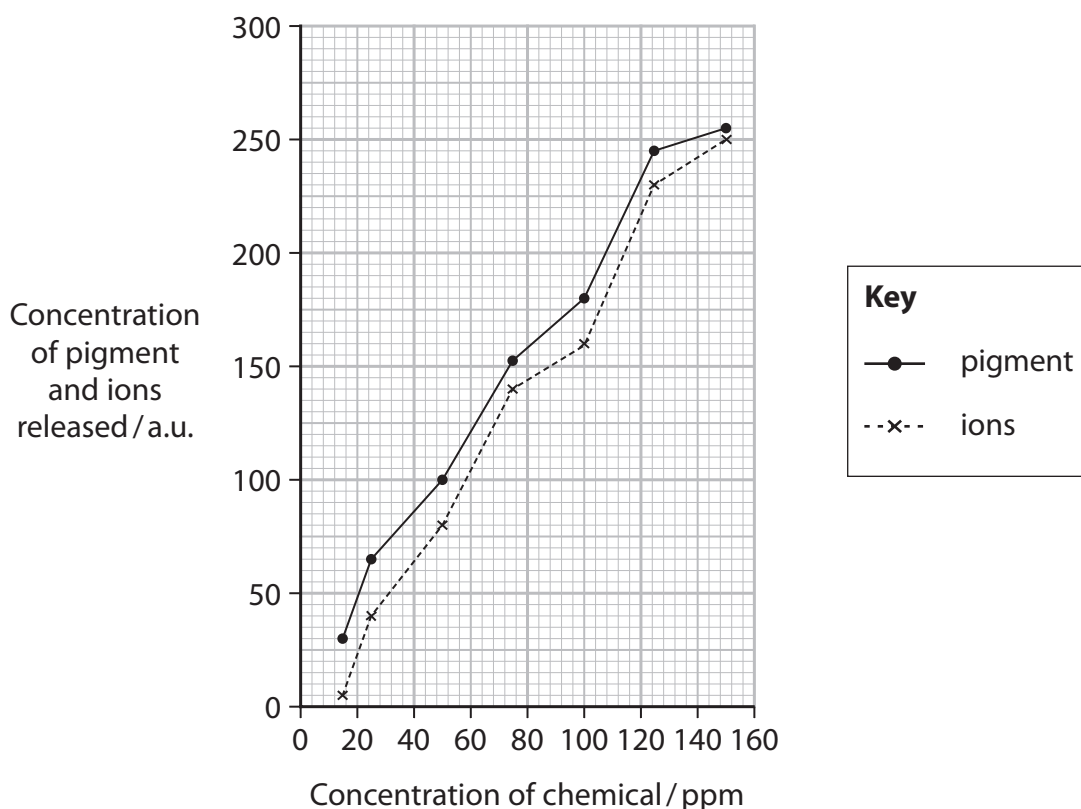
(2)

- 1
- 2
- 3

- (b) A chemical has been shown to affect the permeability of both the tonoplast and the cell membrane.

In one investigation, the permeability of the tonoplast and the cell membrane was measured when the chemical was added to pieces of plant tissue. The concentration of the pigment and ions outside the tissue was used as a measure of the permeability.

The graph shows some data from this investigation.



- (i) Draw a table to show the results for the concentration of ions outside the cells.

(3)



- (ii) Comment on the effect of increasing the concentration of the chemical on the permeability of the membranes to both the pigment and the ions.

(3)

- (iii) Devise a method that could be used to obtain the data for the pigment shown in the graph.

(5)

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- (c) In another investigation, changes in permeability of the membranes were measured over time.

The table shows data for the first three hours.

Time after addition of chemical / hours	Permeability to ions / a.u.	Permeability to pigment / a.u.
0	0.01	0.00
1	0.02	0.00
2	0.03	0.00
3	0.05	0.03

Explain the effect of this chemical on the permeability of the membranes to ions and to pigment over time.

Use the information in the table and in the diagram of a plant cell.

(4)

(Total for Question 2 = 17 marks)

